CS434 DATA SCIENCE **4 Credits / 4 Hours**

Confirmation

This course syllabus was confirmed by Mediterranean Institute of Technology to be valid from the fall semester of 2026.

Position in the Educational System

The course is a single subject course at the South Mediterranean University.

CSE, SWE, REE | Required

Prerequisites

CS280 Introduction to Artificial Intelligence

Course Learning Outcomes

On successful completion of the course, the student will be able to:

Knowledge and Understanding

CLO1: explain the foundations of data science, distinguish data science from data analytics, machine learning, and artificial intelligence, and describe the data science lifecycle, CLO2: explain data preprocessing and descriptive analysis techniques used to prepare and summarize data,

Skills and Abilities

CLO3: perform data gathering, preprocessing and exploratory data analysis to identify patterns, relationships, anomalies, and data-quality issues,

CLO4: create and interpret effective data visualizations to communicate findings and support decision-making,

CLO5: apply feature engineering techniques to transform and represent data for analytical and modeling tasks,

Judgement and Approach

CLO6: develop and evaluate descriptive and predictive models for decision support using appropriate data-driven methods.

CLO7: integrate data gathering, preprocessing, analysis, visualization, modeling, and big data tools in an end-to-end data science project and communicate the results.

CLO8: evaluate the ethical, legal, and privacy implications of data science practice — including data privacy and consent, bias and fairness, and responsible data use — and exercise sound judgement in addressing them.

Course Content

This course introduces the complete data science lifecycle, beginning with the distinction between data science, data analytics, machine learning, and artificial intelligence. Topics include data gathering and manipulation from databases, data warehouses, and web sources; data preprocessing and descriptive analysis; exploratory data analysis; data visualization; feature engineering; descriptive and predictive modeling for decision support; and big data platforms. Labs and a final project integrate these stages into an end-to-end data science workflow.

Moderate Gen AI use

In this course, CS434, students will be informed as to when and how Generative AI (Gen AI) tools will be used in their graded (class) assignments. Students MUST not use gen AI tools during in-class examinations, or assignments unless explicitly permitted and instructed. Students must give credit to gen AI tools whenever used, even if only to generate ideas, by following the suitable citation style guide required by the instructor (MLA/APA/Chicago citation style). The use of

generative AI tools outside of the presented parameters will be considered plagiarism and as violation of SMU Academic Integrity and will be treated as such. Gen AI can provide helpful information but is not always reliable or accurate. If explicitly allowed to use Gen AI, students are fully responsible for the accuracy of the generated/submitted information. The use of generative AI tools outside of the presented parameters will be considered as fabrication and as violation of SMU Academic Integrity and will be treated as such.

Course Implementation and Assessment Language of Instruction: English.

Form of Teaching: The teaching consists of lectures, pop-quizzes, Assignments, in-class exercises, as well as supervision in connection to the exercises

Grades: The final grade will be the weighted average of the following Grades:

15% Quizzes & assignments	20 % Final Project	25% Mid-Term	35% Final

The total points will be calculated and converted into a letter grade on a scale A to F as such: A, A-, B+, B, B-, C+, C, C-, D+, D, D-, F.

Course Evaluation

The course is evaluated through meetings both during and after the course between teachers and student representatives. Further, an anonymous questionnaire is used to ensure written information. The outcome of the evaluations serves to improve the course while running as well as in its next implementation.

Mapping to ABET Student Outcomes

	CLO1	*CLO2*	*CLO3*	*CLO4*	*CLO5*	*CLO6*	*CLO7*	*CLO8*
ASO1	X		X	X		X	X	
ASO6		X	X	X	X		X	X
ASO4								X

ABET Student Outcomes

SO1	Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
SO2	Ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
SO3	Ability to communicate effectively with a range of audiences.
SO4	Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
SO5	Ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO6	Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
SO7	Ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

List of Topics

Week	Topic	Reading	Work due
1	Foundation of Data Science (DS vs DA vs ML vs AI; Data Science Lifecycle)	Material Provided	Lab 1
2	Data Gathering: Types, Sources & Manipulation	Material Provided	Lab 2
3	Data Preprocessing & Descriptive Analysis	Material Provided	Lab 3
4	Exploratory Data Analysis	Material Provided	Lab 4
5	Data Visualization	Material Provided	Lab 5
6	Feature Engineering	Material Provided	Lab 6
7	***Mid-Term Exam***		***Mid-Term Exam***
8	Descriptive Modeling for Decision Support	Material Provided	Lab 7
9	Predictive Modeling for Decision Support	Material Provided	Lab 8
10-11	Integrated Data Science Project Development (application of Weeks 1-9)	Material Provided	Project / Lab
12-13	Big Data Platforms	Material Provided	Lab 9 / Project
14	Project Presentation	Material Provided	Final Project
15	***Final Exam***		***Final Exam***

Course Literature

- VanderPlas, J. (2022). Python Data Science Handbook (2nd ed.). O'Reilly Media.
- Provost, F., & Fawcett, T. (2013). Data Science for Business. O'Reilly Media. Relevant scientific articles, online resources, datasets, and videos will also be provided.
- Larose, D.T. & Larose, C.D. (2015). Data Mining and Predictive Analytics, 2nd ed. Wiley.